PERMUTATIONS AND COMBINATIONS

PERMUTATIONS		CIRCLE PERMUTATIONS	PERMUTATIONS WITH REPETITION
$P(n,r)=\frac{n!}{(n-r)!}$		$(n-1)!$	$\binom{n}{n_1,n_2,\ldots\ldots n_k}=\frac{n!}{n_1!\,n_2!\ldots\ldots n_k!}=P(n;n_1,n_2,\ldots\ldots n_k)$
$P(n,n)=P(n,n-1)=n!$ $P(n,1)=n$ $P(n,0)=1$	Order is important There is no repetiton		Order is important There is repetiton
COMBINATIONS (BINOMIAL COEFFICENT)		**PACSAL RULE**	**COMBINATIONS WITH REPETITION**
$C(n,r)=\binom{n}{r}=\frac{n!}{r!\,(n-r)!}=\frac{P(n,r)}{r!}$		$\binom{n+1}{r}=\binom{n}{r-1}+\binom{n}{r}$	$\binom{n+k-1}{k}=\binom{n+k-1}{n-1}$
$C(n,n)=C(n,0)=C(0,0)=1$ $C(n,1)=n$ $C(n,r)=C(n,n-r)$ $C(n,r)=0,\ \ for\ r>n\ \ or\ \ r<0$	Order is not important There is no repetiton $\binom{-r}{k}=(-1)^k\binom{r+k-1}{k}$		Order is not important There is repetiton

BINOMIAL EXPANSION	BINOMIAL EXPANSION
$(a+b)^n=\sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r=\binom{n}{0}a^nb^0+\binom{n}{1}a^{n-1}b^1+\cdots+\binom{n}{n}a^0b^n=a^n+na^{n-1}b+\frac{n(n-1)}{1.2}a^{n-2}b^2\ldots nab^{n-1}+b^n$	$(x_1+x_1+\cdots+x_k)^n=\sum_{n_1,n_2,\ldots\ldots n_k}^{n}\binom{n}{n_1,n_2,\ldots\ldots n_k}{x_1}^{n_1}{x_2}^{n_2}\ldots\ldots{x_k}^{n_k}\,,\qquad n=n_1+n_1+\cdots+n_k$
	$Number\ of\ terms=\binom{n+k-1}{n}$

MARKOV INEQUALITY	CHEBYSHEV INEQUALITY			DIRAC DELTA FUNCTION and UNIT STEP FUNCTION		
$P(X\geq c)\leq\frac{E(X)}{c}$	$P(\lvert X-\mu\rvert<k\sigma)\geq 1-\frac{1}{k^2}$	$P(\lvert X-\mu\rvert\geq k\sigma)\geq\frac{1}{k^2}$	$P(\lvert X-\mu\rvert\geq k)\geq\frac{\sigma^2}{k^2}$	$\delta(x-x_0)=\begin{cases}\infty, & x=x_0\\ 0, & x\neq x_0\end{cases}$	$\delta(x-x_0)=\frac{d}{dx}u(x-x_0)$	$u(x-x_0)=\begin{cases}1, & if\ \ x\geq x_0\\ 0, & if\ \ x<x_0\end{cases}$
				$\int_{x_0-\varepsilon}^{x_0+\varepsilon}\delta(x-x_0)dx=1$	$\int_{x_0-\varepsilon}^{x_0+\varepsilon}g(x)\ \delta(x-x_0)dx=\int_{-\infty}^{\infty}g(x)\ \delta(x-x_0)dx=g(x_0)$	

PROBABILITY

CONVENTIONAL DEFINITION	FREQUENCY DEFINITION	SUBJECTIVE DEFINITION
$p(A) = \dfrac{n(A)}{n(S)}$	$p(A) = \lim\limits_{N\to\infty} \dfrac{n}{N}$	$\dfrac{p(A)}{1-p(A)} = \dfrac{x}{y} \implies p(A) = \dfrac{x}{x+y}$

PROBABILITY OF UNION	BOOLE INEQUALITY	BONFERONI INEQUALITY
$P(A_1 \cup A_2 \cup \ldots \cup A_n) = \sum\limits_{i=1}^{n} P(A_i) - \sum\limits_{1\le i<j\le n} P(A_i \cap A_j) + \sum\limits_{1\le i<j<k\le n} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n-1} P(A_1 \cap A_2 \cap \ldots \cap A_n)$	$P(A_1 \cup A_2) \le P(A_1) + P(A_2)$	$P(A_1 \cap A_2) \ge P(A_1) + P(A_2) - 1$
	$P\left(\bigcup\limits_{i=1}^{n} A_i\right) \le \sum\limits_{i=1}^{n} P(A_i)$	$P\left(\bigcap\limits_{i=1}^{n} A_i\right) \ge \sum\limits_{i=1}^{n} P(A_i) - 1$

CONDITIONAL PROBABILITY	DISJOINT EVENTS	INDEPENDENT EVENTS	BAYES' THEOREM
$P(A\mid B) = \dfrac{P(A \cap B)}{P(B)}, \quad P(B) > 0$	$\sum\limits_{i=1}^{n} P(B_i) = 1$	*A and B are independent if and only if;* $P(A\mid B) = P(A) \quad or \quad P(B\mid A) = P(B)$	$P(B_r\mid A) = \dfrac{P(B_r).P(A\mid B_r)}{\sum\limits_{i=1}^{n} P(B_i)P(A\mid B_i)}$
$P(A\mid A) = 1$ $P(A_1 \cap A_2) = P(A_1\mid A_2)\cdot P(A_2) = P(A_2\mid A_1)\cdot P(A_1)$ $P\left(\bigcap\limits_{i=1}^{n} A_i\right) = P(A_1)\cdot P(A_2\mid A_1)\cdot P(A_3\mid A_1 A_1 \cap A_2)\cdot \ldots \cdot P\left(A_n \mid \left(\bigcap\limits_{i=1}^{n-1} A_i\right)\right)$	$P(A) = \sum\limits_{i=1}^{n} P(B_i \cap A) = \sum\limits_{i=1}^{n} P(B_i) P(A\mid B_i)$	$P(A \cap B) = P(A)\cdot P(B)$ $P\left(\bigcap\limits_{i=1}^{n} A_i\right) = \prod\limits_{i=2}^{n} P(A_i)$	

DISCRETE RANDOM VARIABLES

PROBABILITY FUNCTIONS (PROBABILITY DISTRIBUTION)	CUMULATIVE DISTRIBUTION FUNCTION	EXPECTED VALUE
$f(x_i) = P(X = x_i)$	$F(x) = \sum_{u \leq x} f(u) \quad for - \infty < x < \infty$	$E(x) = \mu_x = \sum_{i=1}^{n} x_i f(x_i)$
$\sum_{x} f(x_i) = 1$	$P(X > a) = 1 - P(X \leq a) = 1 - F(a)$ $P(a < X \leq b) = F(b) - F(a)$	$E\big(g(x)\big) = \mu_{g(x)} = \sum_{i=1}^{n} g(x_i) f(x_i)$
VARIANCE AND STANDARD DEVIATION	**MOMENT**	**MOMENT GENERATION FUNCTION**
$V(X) = Var(X) = \sigma_X^2 = E[(X-\mu)^2] = \sum_{x} (X-\mu)^2 f(x)$	$m_r = E[(X)^r] = \sum_{i=1}^{n} (x_i)^r f(x_i)$	$M_x(t) = E[e^{tX}] = \sum_{i=1}^{k} e^{x_i t}.f(x_i)$
	$\mu_r = E[(X-a)^r] = \sum_{i=1}^{n} (x_i - a)^r f(x_i)$	$M_x(t) = 1 + tm_1 + \frac{t^2}{2!} m_2 + \frac{t^3}{3!} m_3 + \cdots + \frac{t^r}{r!} m_r + \cdots$

CONTINUOUS RANDOM VARIABLES

PROBABILITY FUNCTIONS	CUMULATIVE DISTRIBUTION FUNCTION	EXPECTED VALUE
$f(x) \geq 0, \quad P(a \leq X \leq b) = \int_{a}^{b} f(x)dx$	$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\,du \quad for - \infty < x < \infty$	$E(x) = \mu_x =$
$\int_{-\infty}^{\infty} f(x)\,dx = 1$; $P(X = a) = 0$ $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$	$\lim_{x \to -\infty} F(x) = F(-\infty) = 0$ $\lim_{x \to \infty} F(x) = F(\infty) = 1$ $F'(x) = f(x)$ $P(X > a) = 1 - F(a)$ $P(a < X \leq b) = F(b) - F(a)$	$E\big(g(x)\big) = \mu_{g(x)} = \int_{-\infty}^{\infty} g(x) f(x)\,dx$
VARİANCE AND STANDARD DEVIATION	**MOMENT**	**MOMENT GENERATION FUNCTION**
$V(X) = Var(X) = \sigma_X^2 = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (X-\mu)^2 f(x)dx$	$m_r = E[(X)^r] = \int_{-\infty}^{\infty} (x)^r f(x)\,dx$	$M_x(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{xt}.f(x)dx$
	$\mu_r = E[(X-a)^r] = \int_{-\infty}^{\infty} (x-a)^r f(x)\,dx$	

MIXED RANDOM VARIABLES

GENERALIZED PROBABILITY FUNCTIONS

$$f_G(x) = \frac{dF(x)}{dx} = \sum_{i=1}^{k} f(x_i)\delta(x - x_i)$$

$$\sum_{x} f(x_i) = 1 \qquad \int_{-\infty}^{\infty} f_G(x)dx = 1$$

CUMULATIVE DISTRIBUTION FUNCTION

$$F(x) = \sum_{i=1}^{k} f(x_i)u(x - x_i)$$

EXPECTED VALUE

$$E(x) = \mu_x = \sum_{i=1}^{n} x_i f(x_i) + \int_{-\infty}^{\infty} xg(x)\,dx$$

$$E\big(g(x)\big) = \sum_{i=1}^{n} g(x_i)f(x_i)$$

VARIANCE AND STANDARD DEVIATION

$$Var(X) = \int_{-\infty}^{\infty} (x - \mu_x)^2 f_G(x)dx = \sum_{i=1}^{n}(x_i - \mu_x)^2 f(x_i) + \int_{-\infty}^{\infty}(x - \mu_x)^2 f(x)\,dx$$

PROPERTIES OF EXPECTED VALUE

- $E(c) = c$
- $E(X + c) = c$
- $E(cX) = cE(X)$
- $x \geq 0 \quad \Longrightarrow \quad E(X) \geq c$
- $x \leq y \quad \Longrightarrow \quad E(X) \leq E(Y)$
- $E(X + Y) = E(X) + E(Y)$
- $|E(X)| \leq E(|X|)$
- X and Y are independent $\quad \Longrightarrow \quad E(XY) = E(X)E(Y)$
- $E(X^2) \neq [E(X)]^2$
- $E(aX + b) = aE(X) + b$

PROPERTIES OF VARIANCE

- $Var(X) = E(x^2) - [E(x)]^2$
- $Var(c) = 0$
- $Var(X + c) = Var(X)$
- $Var(cX) = c^2 Var(X)$
- $Var(aX + b) = a^2 Var(X)$
- $\sigma_X = \sqrt{Var(X)}$

PROPERTIES OF MOMENT

- $m_O = 1$
- $m_1 = E(x) = \mu$
- If $a = \mu$ then;

  $\mu_2 = E[(X - \mu)^2] = Var(X)$

  $\mu_1 = 0$

  $\mu_2 = m_2 - {m_1}^2$

  $\mu_3 = m_3 - 3m_1 m_2 + 2{m_1}^3$

PROPERTIES OF MOMENT GENERATION FUNCTION

- $M_x'(t = 0) = m_1 = E(x) = \mu$
- $M_x''(t = 0) = E(X^2)$
- $M_x^{(n)}(t = 0) = E(X^n)$
- $Var(X) = E(x^2) - [E(x)]^2 = M_x''(0) - [M_x'(0)]^2$
- $M_{X+a}(t) = E\left[e^{(X+a)t}\right] = e^{at}M_X(t)$
- $M_{bX}(t) = E\left[e^{(bX)t}\right] = M_X(bt)$
- $M_Y(t) = M_{aX+b}(t) = e^{bt}M_X(at)$
- $M_{\frac{X+a}{b}}(t) = e^{\frac{a}{b}t}M_X\left(\frac{t}{b}\right)$

PROPERTIES OF COVARIANCE

- $Cov(X,Y) = E\big(X - E(X)\big)\,E\big(Y - E(Y)\big) = E(XY) - E(X)E(Y) = E(XY) - \mu_X\mu_Y$
- $Cov(X,Y) = Cov(Y,X)$
- $Cov(X,X) = Var(X) = {\sigma_X}^2$
- If X and Y are independent $\quad \Longrightarrow \quad {\mu_{1,1}}' - \mu_X\mu_Y = E(X)E(Y) - E(X)E(Y) = 0$
- $Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X,Y)$

PROPERTIES OF CORRELATION

- $-1 \leq \rho_{XY} \leq 1$
- No lineer correlation: $\rho_{XY} = 0$
- Inverse linear correlation: $\rho_{XY} < 0$
- Linear correlation: $\rho_{XY} > 0$
- Perfect inverse linear correlation: $\rho_{XY} = -1$
- Perfect linear correlation: $\rho_{XY} = 1$
- $\rho_{XY} = \rho_{YX}$

PROPERTIES OF JOINT EXPECTED VALUE

$$E(XY) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xy.f(x,y)\,dxdy$$

$$E(X) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} xf(x,y)\,dxdy = \int_{-\infty}^{+\infty} xf(x)\,dx = \mu_x$$

$$E(Y) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} yf(x,y)\,dxdy = \int_{-\infty}^{+\infty} yf(y)\,dy = \mu_y$$

JOINT DISCRETE (DISTRIBUTION) MULTIVARIABLES

JOINT PROBABILITY DISTRIBUTION

$$f(x,y) = P(X = x, Y = y)$$

$$\sum_x \sum_y f(x,y) = 1$$

$$f(X_1, X_2, \dots, X_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

JOINT (CUMULATIVE) DISTRIBUTION FUNCTION

$$F(x,y) = P(X \le x, Y \le y) = \sum_{s \le x} \sum_{t \le y} f(s,t)$$

$$F(X_1, X_2, \dots, X_n) = P(X_1 \le x_1, X_2 \le x_2, \dots, X_n \le x_n)$$

MARGINAL DISTRIBUTION

$$g(x) = \sum_y f(x,y) \qquad h(y) = \sum_x f(x,y)$$

$$g(x_1) = \sum_{x_2} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

$$m(x_1, x_2, x_3) = \sum_{x_4} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

CONDITINAL DISTRIBUTION

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{f(x,y)}{h(y)}$$

$$f(x \mid y) = \frac{f(x,y)}{h(y)} \quad h(y) \neq 0 \qquad w(y \mid x) = \frac{f(x,y)}{g(x)} \quad g(x) \neq 0$$

$$p(x_3 \mid x_1, x_2, x_4) = \frac{f(x_1, x_2, x_3, x_4)}{g(x_1, x_2, x_4)} \quad g(x_1, x_2, x_4) \neq 0$$

$$q(x_2, x_4 \mid x_1, x_3) = \frac{f(x_1, x_2, x_3, x_4)}{m(x_1, x_3)} \quad m(x_1, x_3) \neq 0$$

$$r(x_2, x_3, x_4 \mid x_1) = \frac{f(x_1, x_2, x_3, x_4)}{b(x_1)} \quad b(x_1) \neq 0$$

INDEPENDENCE

$$f(x_1, x_2, \dots, x_n) = f_1(x_1) \cdot f_2(x_2) \cdot \dots \cdot f_n(x_n)$$

$$\mathrm{P}(x \mid y) = \mathrm{P}(x), \quad \mathrm{P}(y) \neq 0 \qquad \mathrm{P}(y \mid x) = \mathrm{P}(y), \quad \mathrm{P}(x) \neq 0$$

$$\mathrm{P}(x \mid y) = \frac{\mathrm{P}(x, y)}{\mathrm{P}(y)} = \frac{\mathrm{P}(x)\mathrm{P}(y)}{\mathrm{P}(y)} = \mathrm{P}(x)$$

EXPECTED VALUE

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) \cdot f(x,y)$$

$$E\left[\sum_{i=1}^{n} c_i g_i(X_1, X_2, \dots, X_k)\right] = \sum_{i=1}^{n} c_i E[g_i(X_1, X_2, \dots, X_k)]$$

CONDITIONAL EXPECTATION

$$E[u(X) \mid y] = \sum_x u(x) \cdot f(x \mid y)$$

COVARIANCE

$$\sigma_{XY} = Cov(X,Y) = C(X,Y) = \mu_{1,1}{}' - \mu_X \mu_Y$$

MOMENTS OF LINEAR COMBINATIONS

$$If \quad Y = Y_1 = \sum_{i=1}^{n} a_i X_i \quad and \quad Y_2 = \sum_{i=1}^{n} b_i X_i, \quad then;$$

$$E(Y) = \sum_{i=1}^{n} a_i E(X_i) \qquad \mathrm{var}(Y) = \sum_{i=1}^{n} a_i^2 \cdot \mathrm{var}(X_i) + 2 \sum_{i<j} \sum a_i a_j \cdot \mathrm{cov}(X_i X_j)$$

$$If\ X_1, X_2, \dots, X_n\ are\ independent, then;\ Var(Y) = \sum_{i=1}^{n} a_i{}^2 Var(X_i)$$

$$\mathrm{cov}(Y_1, Y_2) = \sum_{i=1}^{n} a_i b_i \cdot \mathrm{var}(X_i) + \sum_{i<j} \sum (a_i b_j + a_j b_i) \cdot \mathrm{cov}(X_i, X_j)$$

$$If\ X_1, X_2, \dots, X_n\ are\ independent, then;\ Var(Y_1, Y_2) = \sum_{i=1}^{n} a_i b_i Var(X_i)$$

CORRELATION

$$\mathrm{corr}(X,Y) = \rho_{xy} = \frac{\mathrm{Cov}(X,Y)}{\sigma_x \sigma_y} = \frac{E\left[(X - \mu_X)(Y - \mu_Y)\right]}{\sqrt{E\left[(X - \mu_X)^2\right]}\sqrt{E\left[(Y - \mu_Y)^2\right]}}$$

$If\ Y = aX + b\ then;$

$$\rho_{xy} = \begin{cases} -1, & a < 0 \\ 0, & a = 0 \\ 1, & a > 0 \end{cases}$$

PRODUCT MOMENTS

$$\mu'_{r,s} = E(X^r Y^s) = \sum_x \sum_y x^r y^s \cdot f(x,y)$$

$$\mu_{r,s} = E[(X - \mu_X)^r (Y - \mu_Y)^s]$$

$$= \sum_x \sum_y (x - \mu_X)^r (y - \mu_Y)^s \cdot f(x,y)$$

JOINT CONTINUOUS (DENSITY) MULTIVARIABLES

JOINT PROBABILITY DENSITY FUNCTIONS (PDF)

$$P[(X,Y)\in A] = P(a<X<b,\ c<Y<d) = \iint_A f(x,y)dxdy$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy = 1$$

$$f(x_1,x_2,\dots,x_n) = \frac{\partial^n}{\partial x_1 \partial x_2 \dots \partial x_n} F(x_1,x_2,\dots,x_n)$$

JOINT DISTRIBUTION FUNCTION

$$F(x,y) = P(X\le x, Y\le y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f(s,t)dsdt$$

$$f(x,y) = F_{xy}(x,y) \qquad F_x(x,y) = \int_{-\infty}^{y} f(x,t)\partial t \qquad F_y(x,y) = \int_{-\infty}^{y} f(s,t)\partial s$$

$$F(X_1,X_2,\dots,X_n) = \int_{-\infty}^{x_n}\dots\int_{-\infty}^{x_2}\int_{-\infty}^{x_1} f(t_1,t_2,\dots,t_n)dt_1dt_2\dots dt_n$$

MARGINAL DENSITY

$$g(x) = \int_{-\infty}^{\infty} f(x,y)dy \qquad h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

$$h(x_2) = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1,x_2,\dots,x_n)\,dx_1\,dx_3\cdots dx_n$$

$$\varphi(x_1,x_n) = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x_1,x_2,\dots,x_n)\,dx_2\,dx_3\cdots dx_{n-1}$$

CONDITINAL DENSITY

$$f(x|y) = \frac{f(x,y)}{h(y)} \qquad h(y)\neq 0 \qquad w(y|x) = \frac{f(x,y)}{g(x)} \qquad g(x)\neq 0$$

$$p(x_3|x_1,x_2,x_4) = \frac{f(x_1,x_2,x_3,x_4)}{g(x_1,x_2,x_4)} \qquad g(x_1,x_2,x_4)\neq 0$$

$$q(x_2,x_4|x_1,x_3) = \frac{f(x_1,x_2,x_3,x_4)}{m(x_1,x_3)} \qquad m(x_1,x_3)\neq 0$$

$$r(x_2,x_3,x_4|x_1) = \frac{f(x_1,x_2,x_3,x_4)}{b(x_1)} \qquad b(x_1)\neq 0$$

INDEPENDENCE

$$f(x_1,x_2,\dots,x_n) = f_1(x_1)\cdot f_2(x_2)\cdot \dots \cdot f_n(x_n)$$

$$P(x|y) = P(x),\quad P(y)\neq 0 \qquad P(y|x) = P(y),\quad P(x)\neq 0$$

$$P(x|y) = \frac{P(x,y)}{P(y)} = \frac{P(x)P(y)}{P(y)} = P(x)$$

EXPECTED VALUE

$$E[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f(x,y)\,dx\,dy$$

$$E\left[\sum_{i=1}^{n} c_i g_i(X_1,X_2,\dots,X_k)\right] = \sum_{i=1}^{n} c_i E[g_i(X_1,X_2,\dots,X_k)]$$

CONDITIONAL EXPECTATION

$$E[(u(X)|y)] = \int_{-\infty}^{\infty} u(x)\cdot f(x|y)dx$$

COVARIANCE

$$\sigma_{XY} = Cov(X,Y) = C(X,Y) = \mu_{1,1}{}' - \mu_X\mu_Y$$

MOMENTS OF LINEAR COMBINATIONS

$$If \quad Y = Y_1 = \sum_{i=1}^{n} a_i X_i \quad and \quad Y_2 = \sum_{i=1}^{n} b_i X_i, \quad then;$$

$$E(Y) = \sum_{i=1}^{n} a_i E(X_i) \qquad \mathrm{var}(Y) = \sum_{i=1}^{n} a_i^2\cdot \mathrm{var}(X_i) + 2\sum\sum_{i<j} a_i a_j \cdot \mathrm{cov}(X_i X_j)$$

$$If\ X_1,X_2,\dots,X_n\ are\ independent, then;\ Var(Y) = \sum_{i=1}^{n} {a_i}^2 Var(X_i)$$

$$\mathrm{cov}(Y_1,Y_2) = \sum_{i=1}^{n} a_i b_i \cdot \mathrm{var}(X_i) + \sum\sum_{i<j}(a_i b_j + a_j b_i)\cdot \mathrm{cov}(X_i,X_j)$$

$$If\ X_1,X_2,\dots,X_n\ are\ independent, then;\ Var(Y_1,Y_2) = \sum_{i=1}^{n} a_i b_i Var(X_i)$$

CORRELATION

$$\mathrm{corr}(X,Y) = \rho_{xy} = \frac{\mathrm{Cov}(X,Y)}{\sigma_x\sigma_y} = \frac{E\left[(X-\mu_X)(Y-\mu_Y)\right]}{\sqrt{E\left[(X-\mu_X)^2\right]}\sqrt{E\left[(Y-\mu_Y)^2\right]}}$$

$If\ Y = aX + b\ then;$

$$\rho_{xy} = \begin{cases} -1, & a<0 \\ 0, & a=0 \\ 1, & a>0 \end{cases}$$

PRODUCT MOMENTS

$$\mu'_{r,s} = E(X^r Y^s) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^r y^s \cdot f(x,y)dxdy$$

$$\mu_{r,s} = E[(X-\mu_X)^r(Y-\mu_Y)^s]$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_X)^r(y-\mu_Y)^s\cdot f(x,y)dxdy$$

SPECIAL PROBABLITY DISTRIBUTIONS

DISCRETE UNIFORM DISTRIBUTION

$$f(x;n)=\frac{1}{n};\quad x=x_1,x_2,\ldots,x_n$$

VARIANCE

$$Var(X)=E(X^2)-[E(X)]^2=\frac{n^2-1}{12}$$

EXPECTED VALUE

$$E(X)=\frac{1}{n}.\frac{n(n+1)}{2}=\frac{n+1}{2}$$

$$E(X^2)=\frac{1}{n}.\frac{n(n+1)(2n+1)}{6}=\frac{(n+1)(2n+1)}{6}$$

CONTINUOUS UNIFORM DISTRIBUTION

$$\boxed{u(x;a,b)=\begin{cases}\frac{1}{b-a}, & a\le x\le b\\ 0, & \text{Elsewhere}\end{cases}}$$

CUMULATIVE PROBABILITY FUNCTION

$$\boxed{F(x)=\begin{cases}0 & x<a\\ \frac{x-a}{b-a} & a\le x\le b\\ 1 & x>b\end{cases}}$$

$$P(X\le x)=F(x)=\int_a^x\frac{1}{b-a}dx=\frac{x-a}{b-a}$$

THE BERNOULLI DISTRIBUTION

$$f(x;p)=p^x(1-p)^{1-x}\quad x=0,1$$

$p+q=1;$

$$f(x;p)=\begin{cases}1-p=q, & x=0\\ p, & x=1\\ 0, & \text{Other cases}\end{cases}$$

i) $f(x;p)=p^x(1-p)^{1-x}\ge 0$

ii) $\sum_{x=0}^{1}f(x;p)=\sum_{x=0}^{1}p^xq^{1-x}=q+p=1$

BERNOULLI EXPECTED VALUE

$$M'(t)=e^t.p$$
$$M'(0)=p$$
$$E(X)=\mu=p$$

$$M''(t)=e^t.p$$
$$M''(0)=p$$
$$E(X^2)=p$$

BERNOULLI MOMENT GENERATING FUNCTION

$$M(t)=E[e^{tX}]=q+e^t.p$$

BERNOULLI VARIANCE

$$Var(X)=E(X^2)-[E(X)]^2=pq$$

BINOMIAL DISTRIBUTION		BINOMIAL EXPECTED VALUE	
$b(x;n,p)=\binom{n}{x}p^x(1-p)^{n-x} \quad x=0,1,2,......,n$ $b(x;n,p)=b(n-x;n,1-p)$	i) $b(x;n,p)\geq 0$ ii) $\sum_{x=0}^{n} b(x;n,p)=\sum_{x=0}^{n}\binom{n}{x}p^x q^{n-x}=(p+q)^n=1$	$E(X)=\mu=E\left[\sum_{i=1}^{n}Y_i\right]$ $M'(t)=n(pe^t+q)^{n-1}.pe^t$ $M'(0)=E(X)=\mu=np$	$M''(t)=n(pe^t+q)^{n-2}.pe^t\left[(n-1)pe^t+(pe^t+q)\right]$ $M''(0)=n^2p^2+npq$ $E\left(\frac{X}{n}\right)=\frac{1}{n}E(X)=\frac{1}{n}np=p$
BINOMIAL MOMENT GENERATING FUNCTION		**BINOMIAL VARIANCE**	
$M(t)=E[e^{tX}]=\sum_{x=0}^{n}\binom{n}{x}(e^tp)^x q^{n-x}$	$M(t)=\sum_{x=0}^{n}\binom{n}{x}r^x q^{n-x}=(r+q)^n=(pe^t+q)^n$	$Var(X)=Var\left[\sum_{i=1}^{n}Y_i\right]$ $Var\left(\frac{X}{n}\right)=\frac{1}{n^2}Var(X)=\frac{pq}{n}$	$Var(X)=E(X^2)-[E(X)]^2$ $=n^2p^2+npq-n^2p^2$ $=npq$

MULTINOMIAL DISTRIBUTION	MULTINOMIAL EXPECTED VALUE
$f(x_1,x_2,....,x_k;n,p_1,p_2,...p_k)=\binom{n}{x_1,x_2,...,x_k}p_1^{x_1}p_2^{x_2}.....p_k^{x_k}$ For all i $x_i\geq 0,\ \ i=1,...,k \quad \sum_{i=1}^{k}x_i=n$ ve $\sum_{i=1}^{k}p_i=1$	$E[X_i]=\mu_i=np_i$
	MULTINOMIAL VARIANCE
	$Var[X_i]=np_iq_i$

GEOMETRIC DISTRIBUTION

$$g(x;p) = q^{x-1}p \qquad x = 1, 2, 3, \ldots$$

$$\sum_{x=1}^{\infty} g(x;p) = \sum_{x=1}^{\infty} q^{x-1}p = p(1+q+q^2+q^3+\ldots.) = p\frac{1}{1-q} = 1$$

GEOMETRIC EXPECTED VALUE

$$M'(t) = \frac{pe^t(1-e^tq)+pqe^{2t}}{(1-e^tq)^2}$$

$$M'(0) = E(X) = \frac{1}{p}$$

$$E(X) = \mu = \sum_{x=1}^{n} x.q^{x-1}.p$$

$$M''(0) = E(X^2) = \frac{2q+p}{p^2}$$

GEOMETRIC MOMENT GENERATING FUNCTION

$$M(t) = E[e^{tX}] = \frac{pe^t}{1-e^tq}$$

GEOMETRIC VARIANCE

$$Var(X) = M''(0) - [M'(0)]^2$$

$$Var(X) = \frac{2q+p}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

EGATIVE BINOMIAL (PASCAL) DISTRIBUTION

$$b^*(x;k,p) = \binom{x-1}{k-1}p^kq^{x-k} \qquad x = k, k+1, k+2, \ldots$$

$$b^*(x;k,p) = \frac{k}{x}.b(x;k,p)$$

$$\sum_{x=k}^{\infty}\binom{x-1}{k-1}p^kq^{x-k} = p^k(1-q)^{-k} = 1$$

NEGATIVE BINOMIAL EXPECTED VALUE

$$E[X_i] = \frac{1}{p}$$

$$E[X] = E[X_1] + E[X_2] + \ldots. + E[X_k] = k.\frac{1}{p}$$

NEGATIVE BINOMIAL VARIANCE

$$Var[X_i] = \frac{q}{p^2}$$

$$Var[X] = Var[X_1] + Var[X_2] + \ldots. + Var[X_k] = \frac{kq}{p^2}$$

HYPERGEOMETRIC DISTRIBUTION

$$h(x;n,N,M) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} \qquad x = 0, 1, 2, \ldots, n_k \qquad n_k = \begin{cases} n, & n \le M \\ M, & n > M \end{cases}$$

$$\sum_{x=0}^{n}\binom{M}{x}\binom{N-M}{n-x} = \binom{N}{n}$$

$$\sum_{x=0}^{n} h(x;n,N,M) = 1$$

HYPERGEOMETRIC EXPECTED VALUE

$$\mu = E(X) = \sum_{x=0}^{n} x.\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} = \frac{Mn}{N}$$

$$E(X) = \mu = \frac{M}{N}.n = p.n$$

$$E(X^2) = \frac{M(M-1)n(n-1)}{N(N-1)} + \frac{Mn}{N}$$

HYPERGEOMETRIC BINOMIAL VARIANCE

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{N-n}{N-1}.n.\frac{M}{N}\left(1-\frac{M}{N}\right)$$

$$Var(X) = npq\frac{N-n}{N-1}$$

MULTIVARIATE HYPERGEOMETRIC DISTRIBUTION		
$f(x_1, x_2, ..., x_k; n, M_1, M_2, ..., M_k) = \dfrac{\binom{M_1}{x_1}.\binom{M_2}{x_2}.....\binom{M_k}{x_k}}{\binom{N}{n}}$	$\sum_{i=1}^{k} x_i = n, \quad \sum_{i=1}^{k} M_i = N$	$\sum_{x=0}^{n} h(x; n, N, M) = 1$

POISSION DISTRIBUTION			POISSION EXPECTED VALUE	
$b(x; n, p) = \binom{n}{x}\left(\frac{\lambda}{n}\right)^x\left(1-\frac{\lambda}{n}\right)^{n-x} = \frac{\lambda^x e^{-\lambda}}{x!}$		$p(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \text{for } x = 0, 1, 2, ...$	$E(X) = \sum_{x=0}^{\infty} x.\frac{e^{-\lambda}\lambda^x}{x!} = \lambda$	$M'(0) = E(X) = \mu = \lambda$ $M''(0) = \lambda(1+\lambda)$
$np = \lambda \quad p = \frac{\lambda}{n}$	$\lim_{n\to\infty}\binom{n}{x}p^x q^{n-x} \to \frac{\lambda^x e^{-\lambda}}{x!}$	$\sum_{x=0}^{\infty}\frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\left(1+\frac{\lambda}{1}+\frac{\lambda^2}{2!}+....+\frac{\lambda^n}{n!}+..\right) = e^{-\lambda}e^{\lambda} = 1$		

POISSION MOMENT GENERATING FUNCTION	POISSION VARIANCE
$M(t) = E[e^{tX}] = \sum_{x=0}^{\infty} e^{tx}\frac{e^{-\lambda}\lambda^x}{x!} = e^{(e^t-1)\lambda}$	$Var(X) = M''(0) - [M'(0)]^2 = \lambda$